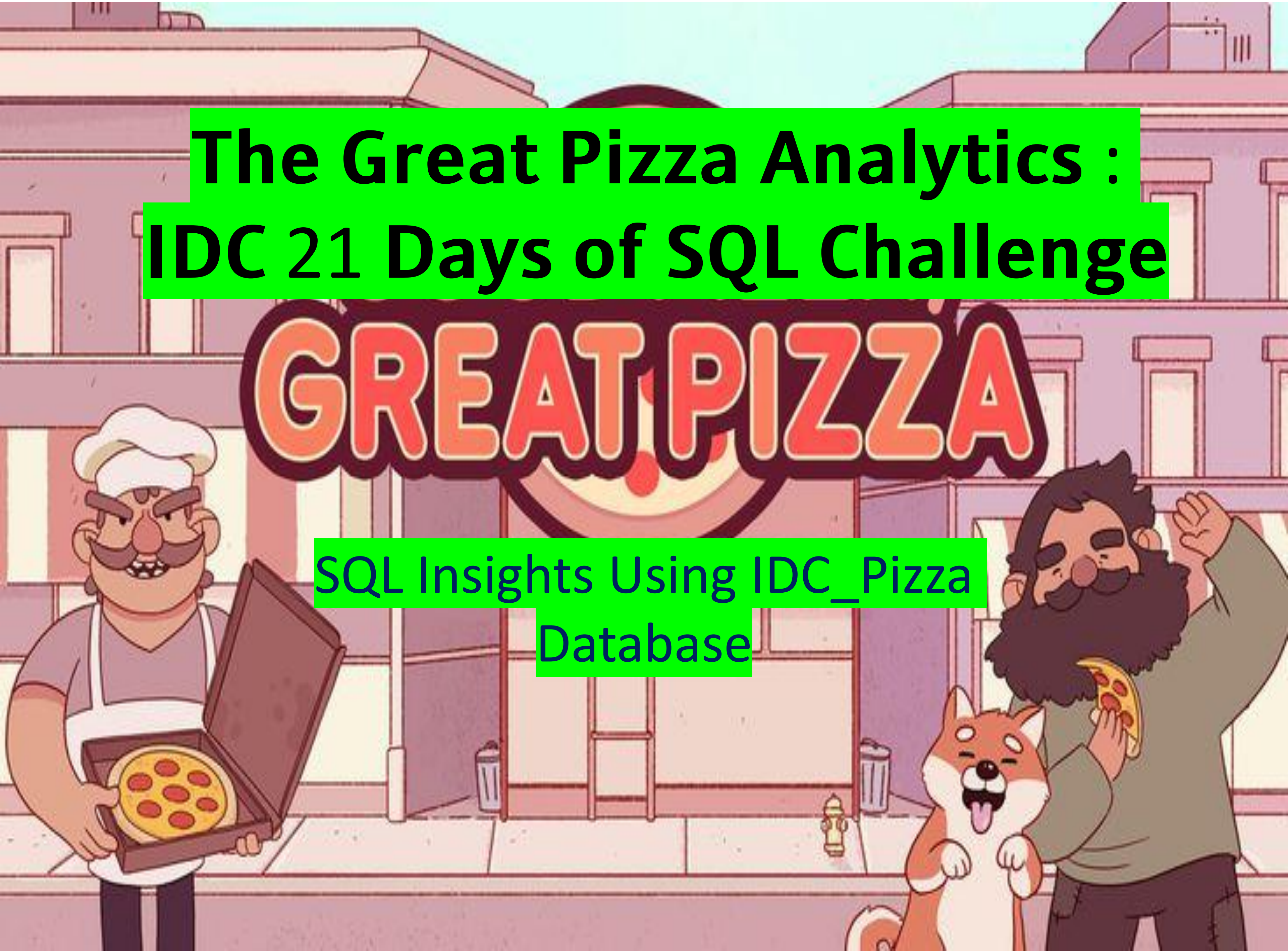
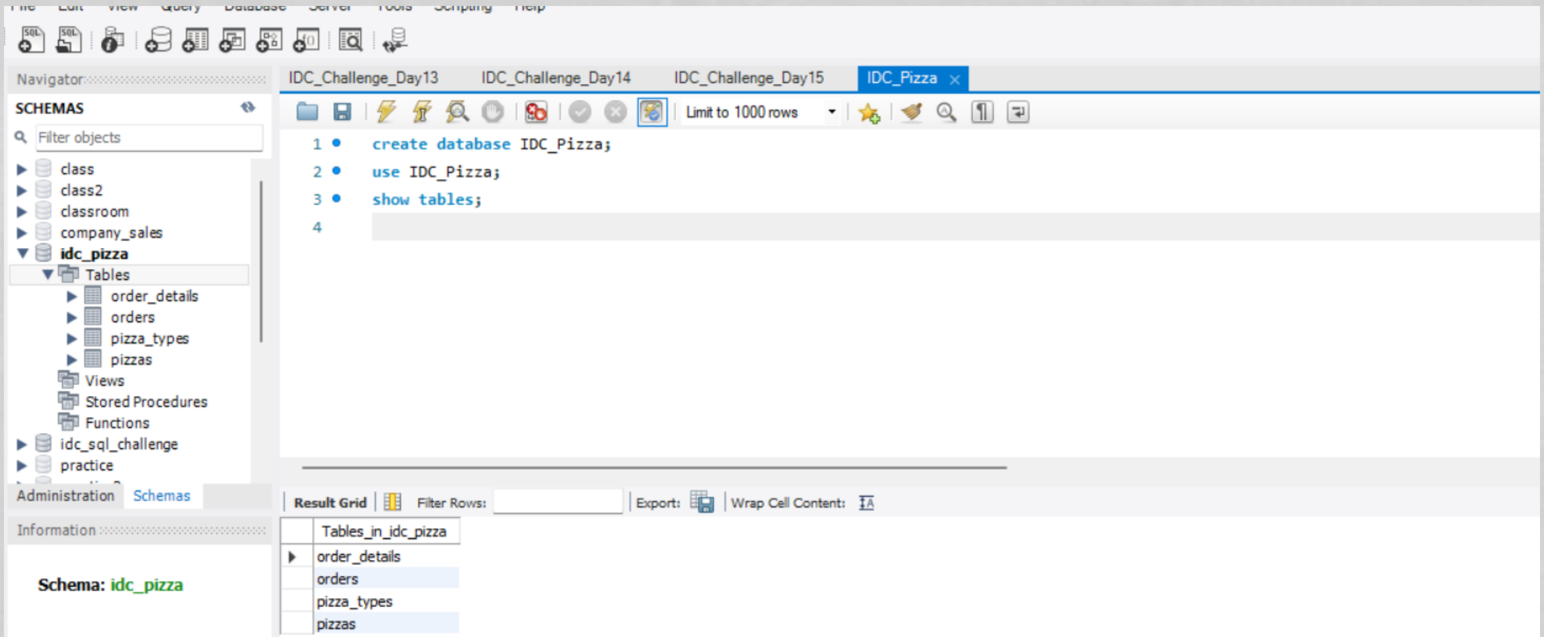
The Great Pizza Analytics : IDC 21 Days of SQL Challenge

SQL Insights Using IDC_Pizza Database

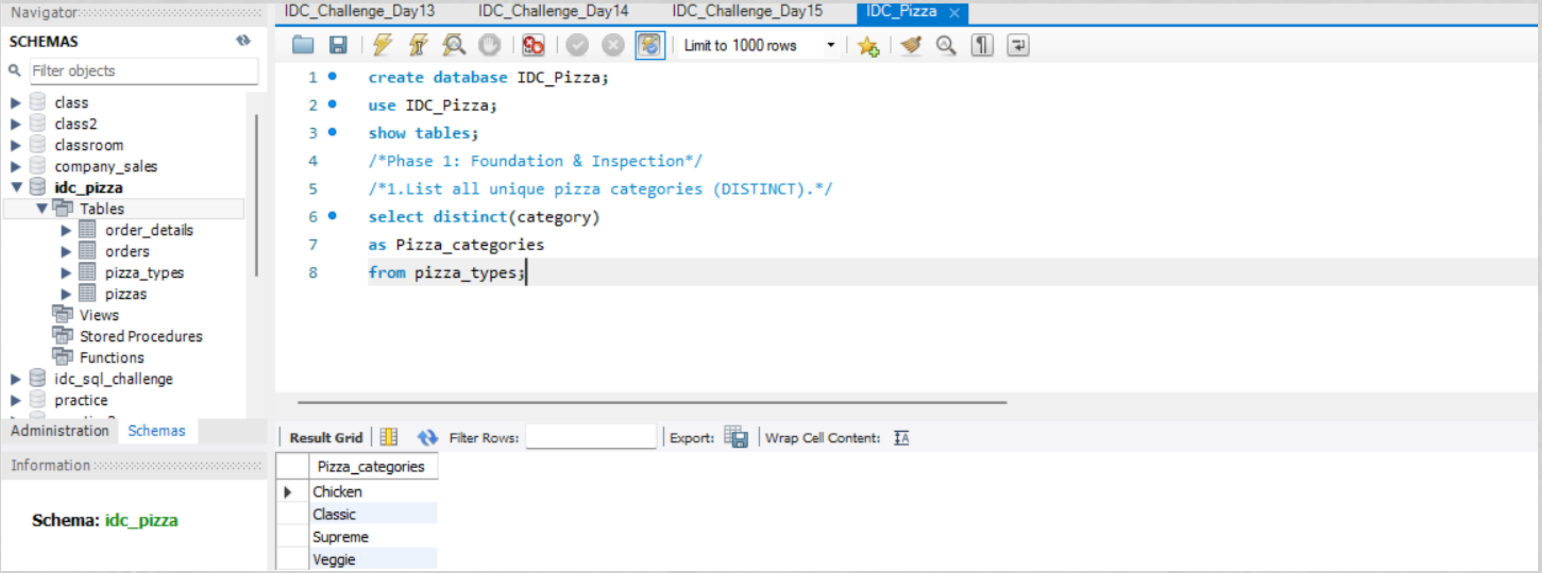


- Topics practiced:
- * Database creation & table design
- * Filtering & operators (WHERE, IN, BETWEEN, LIKE)
- * Aggregations (SUM, AVG, COUNT, GROUP BY, HAVING)
- * Joins (INNER, LEFT, RIGHT, SELF)
- * Data cleaning (DISTINCT, COALESCE, NULL handling)

PIZZA


Navigator
SCHEMAS
Filter objects
class
class2
classroom
company_sales
idc_pizza
Tables
order_details
orders
pizza_types
pizzas
Views
Stored Procedures
Functions
idc_sql_challenge
practice
Administration
Schemas
Information
Schema: idc_pizza
IDC_Challenge_Day13
IDC_Challenge_Day14
IDC_Challenge_Day15
IDC_Pizza
Limit to 1000 rows
1 create database IDC_Pizza;
2 use IDC_Pizza;
3 show tables;
4
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Tables_in_idc_pizza
order_details
orders
pizza_types
pizzas

PIZZA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IDC_Challenge_Day13
IDC_Challenge_Day14
IDC_Challenge_Day15
IDC_Pizza
Limit to 1000 rows
create database IDC_Pizza;
use IDC_Pizza;
show tables;
/*Phase 1: Foundation & Inspection*/
/*1.List all unique pizza categories (DISTINCT).*/
select distinct(category)
as Pizza_categories
from pizza_types;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Pizza_categories
Chicken
Classic
Supreme
Veggie

```sql
as Pizza_categories
from pizza_types;
/*2. Display pizza_type_id, name, and ingredients, replacing NULL ingredients with "Missing Data". Show first 5 rows. */
select pizza_type_id,name,
case when ingredients is null or ingredients = ''
then 'Missing Data'
else ingredients
end as ingredients
from pizza_types
limit 5;
```

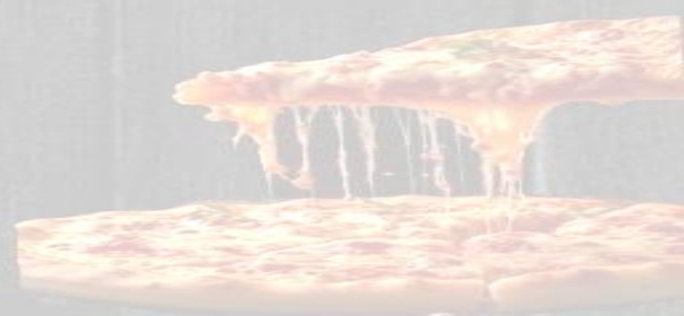
Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

pizza_type_id	name	ingredients
bbq_ckn	The Barbecue Chicken Pizza	Barbecued Chicken, Red Peppers, Green Peppe...
cali_ckn	The California Chicken Pizza	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
ckn_alfredo	The Chicken Alfredo Pizza	Chicken, Red Onions, Red Peppers, Mushrooms...
ckn_pesto	The Chicken Pesto Pizza	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
southw_ckn	The Southwest Chicken Pizza	Chicken, Tomatoes, Red Peppers, Red Onions, ...

```sql
/* 3. Check for pizzas missing a price (IS NULL).*/
select * from pizzas
where price is null;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_id	pizza_type_id	size	price



```sql
/*4. Orders placed on '2015-01-01' (SELECT + WHERE).*/
select order_id,date
from orders
where date = '2015-01-01';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	date
1	2015-01-01
2	2015-01-01
3	2015-01-01
4	2015-01-01
5	2015-01-01

```sql
/*5. List pizzas with price descending.*/
select * from pizzas
order by price desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_id	pizza_type_id	size	price
the_greek_xxl	the_greek	XXL	35.95
the_greek_xl	the_greek	XL	25.5
brie_carre_s	brie_carre	S	23.65
ital_veggie_l	ital_veggie	L	21
spinach_supr_l	spinach_supr	L	20.75
bbq_ckn_l	bbq_ckn	L	20.75
cali_ckn_l	cali_ckn	L	20.75
spicy_ital_l	spicy_ital	L	20.75
ckn_alfredo_l	ckn_alfredo	L	20.75
ckn_pesto_l	ckn_pesto	L	20.75
ital_supr_l	ital_supr	L	20.75

```sql
/*6. Pizzas sold in sizes 'L' or 'XL'.*/
select pizza_id,pizza_type_id,size
from pizzas
where size in ('L','XL');
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_id	pizza_type_id	size
ckn_alfredo_l	ckn_alfredo	L
ckn_pesto_l	ckn_pesto	L
southw_ckn_l	southw_ckn	L
thai_ckn_l	thai_ckn	L
big_meat_l	big_meat	L
classic_dlx_l	classic_dlx	L
hawaiian_l	hawaiian	L
ital_cpcllo_l	ital_cpcllo	L
napolitana_l	napolitana	L
pep_msh_pe...	pep_msh_pep	L
pepperoni_l	pepperoni	L
the_greek_l	the_greek	L
the_greek_xl	the_greek	XL
calabrese_l	calabrese	L
ital_supr_l	ital_supr	L
peppr_salami_l	peppr_salami	L

```sql
/*7. Pizzas priced between $15.00 and $17.00.*/
select * from pizzas
where price between 15.00 and 17.00;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_id	pizza_type_id	size	price
bbq_ckn_m	bbq_ckn	M	16.75
cali_ckn_m	cali_ckn	M	16.75
ckn_alfredo_m	ckn_alfredo	M	16.75
ckn_pesto_m	ckn_pesto	M	16.75
southw_ckn_m	southw_ckn	M	16.75
thai_ckn_m	thai_ckn	M	16.75
big_meat_m	big_meat	M	16
classic_dlx_m	classic_dlx	M	16
hawaiian_l	hawaiian	L	16.5
ital_cpcllo_m	ital_cpcllo	M	16
napolitana_m	napolitana	M	16
pepperoni_l	pepperoni	L	15.25
the_greek_m	the_greek	M	16
calabrese_m	calabrese	M	16.25
ital_supr_m	ital_supr	M	16.5

```
/*8. Pizzas with "Chicken" in the name.*/
select * from pizza_types
where name like '%Chicken%';
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_type_id	name	category	ingredients
bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppe...
cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms...
ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, ...
thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, T...

```sql
/*9. Orders on '2015-02-15' or placed after 8 PM.*/
select * from orders
where date = '2015-02-15' or time > '20:00:00';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	date	time
60	2015-01-01	20:05:16
61	2015-01-01	20:08:43
62	2015-01-01	20:50:16
63	2015-01-01	20:51:42
64	2015-01-01	20:52:08
65	2015-01-01	21:16:00
66	2015-01-01	21:47:55
67	2015-01-01	22:03:40
68	2015-01-01	22:07:32
69	2015-01-01	22:12:13
123	2015-01-02	20:12:09
124	2015-01-02	20:12:34
125	2015-01-02	20:31:06
126	2015-01-02	20:53:42
127	2015-01-02	20:58:23
128	2015-01-02	21:05:06

```sql
/*10. Total quantity of pizzas sold (SUM).*/
select sum(quantity) as Total_quantity_sold
from order_details;
```

Result Grid | Filter Rows: | Export: | Wrap Cel

Total_quantity_sold
49574

```sql
/*11. Average pizza price (AVG).*/
select round(avg(price),2) as Average_price
from pizzas;
```

Average_price
16.44

```sql
/*12. Total order value per order (JOIN, SUM, GROUP BY).*/
select o.order_id, round(sum(p.price * o.quantity),2)
as Total_order_value
from order_details o
join pizzas p
on o.pizza_id = p.pizza_id
group by o.order_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	Total_order_value
4529	38.25
4517	12.75
4477	66.25
4397	42.75
4390	33.5
4374	25.5
4300	49.75
4297	58.25
4294	33
4284	33.5
4219	29.25
4186	237.15
4131	106.25
4125	12.75
4076	27.5
4014	219.25

```sql
/*13. Total quantity sold per pizza category (JOIN, GROUP BY).*/
select pt.category, sum(o.quantity)
as Total_quantity
from order_details o
join pizzas p
on o.pizza_id = p.pizza_id
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	Total_quantity
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

```sql
/*14. Categories with more than 5,000 pizzas sold (HAVING).*/
select pt.category, sum(o.quantity)
as Total_quantity
from order_details o
join pizzas p
on o.pizza_id = p.pizza_id
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.category
having sum(o.quantity) > 5000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	Total_quantity
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

```sql
/* 15. Pizzas never ordered (LEFT/RIGHT JOIN).*/
select p.pizza_id,pt.name
from pizzas p
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
left join order_details o
on p.pizza_id = o.pizza_id
where o.pizza_id is null;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

pizza_id	name
big_meat_m	The Big Meat Pizza
big_meat_l	The Big Meat Pizza
five_cheese_s	The Five Cheese Pizza
five_cheese_m	The Five Cheese Pizza
four_cheese_s	The Four Cheese Pizza

```sql
/*16. Price differences between different sizes of the same pizza (SELF JOIN).*/
select p1.pizza_type_id,p1.size,p2.size,
round(abs(p1.price-p2.price),2) as price_difference
from pizzas p1
join pizzas p2
on p1.pizza_type_id = p2.pizza_type_id
where p1.size <> p2.size and p1.size < p2.size
order by price_difference desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pizza_type_id	size	size	price_difference
the_greek	S	XXL	23.95
the_greek	M	XXL	19.95
the_greek	L	XXL	15.45
the_greek	S	XL	13.5
the_greek	XL	XXL	10.45
the_greek	M	XL	9.5
classic_dlx	L	S	8.5
big_meat	L	S	8.5
ital_cpcllo	L	S	8.5
napolitana	L	S	8.5
the_greek	L	S	8.5
green_garden	L	S	8.25
mexicana	L	S	8.25
mediterraneo	L	S	8.25

Key Business Insights

📊 Sales Performance

•Total pizzas sold: **49,574**

•Average price: **$16.44**

•Classic category leads in volume

💡 Strategic Recommendations

•Promote underperforming items (Big Meat, Five Cheese)

•Leverage late-night ordering patterns (>8PM)

•Focus on $15-17 price range for combo deals

CONCLUSION

• The IDC Pizza Analytics Challenge demonstrates how SQL can turn raw operational data into meaningful insights.
By analyzing menu structure, customer behavior, pricing, and sales, I was able to derive strategies that directly support:

• Better menu management
• Stronger customer engagement
• Optimized pricing
• Improved operational efficiency

This project showcases the impact of data analytics in driving business growth